Interactive Real-Time Motion Tracking based System for Learning Sri Lankan udarata (Kandyan) Traditional Dance

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Udarata (Kandyan) dance is a significant form of Sri Lankan intangible cultural heritage characterized by complex rhythmic movements, intricate footwork, and precise body postures. Traditional training follows a teacher-centered model, where students rely on direct supervision and corrective feedback from expert gurus. However, during independent practice sessions, learners lack real-time guidance, leading to persistent posture errors and movement inaccuracies that may affect both technical quality and cultural authenticity.

Although pose estimation and motion analysis technologies have been successfully applied to ballet and Indian classical dance forms, comprehensive and interactive systems tailored specifically to Udarata dance remain limited. Existing Sri Lankan research primarily focuses on static pose detection rather than dynamic sequence evaluation with real-time corrective feedback.

This research proposes an interactive real-time motion tracking system using MediaPipe pose estimation to analyze Udarata dance movements through standard webcam input. The system extracts 3D body landmarks, applies normalization techniques, and compares student performance with expert reference data using Dynamic Time Warping (DTW). It provides instant visual feedback through personalized guidance to correct posture and movement deviations.

The expected outcomes include improved independent learning accuracy, reduced dependency on constant instructor supervision, and the creation of a digital archive of expert dance movements. Socially, the system contributes to the preservation of Sri Lankan traditional dance heritage. Commercially, it has potential applications in dance academies, cultural institutions, and online learning platforms.

Keywords: Udarata Dance, MediaPipe, Pose Estimation, Motion Tracking, Cultural Preservation

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ABBREVIATION	FULL FORM
AI	Artificial Intelligence
Bi-LSTM	Bi-directional Long Short-Term Memory
CSV	Comma-Separated Values
DSR	Design Science Research
DTW	Dynamic Time Warping
FPS	Frames Per Second
GPU	Graphics Processing Unit
LKR	Sri Lankan Rupee
LSTM	Long Short-Term Memory
PAP	Position Accuracy Percentage
RAP	Rotation Accuracy Percentage
RGB	Red Green Blue (Color Model)
SLIIT	Sri Lanka Institute of Information Technology
UI	User Interface
VR	Virtual Reality
WBS	Work Breakdown Structure

1. INTRODUCTION

1.1 Background

Udarata (Kandyan) dance is a classical dance form that originates from the Kandyan Kingdom of Sri Lanka. It is renowned for its complex rhythmic movements, graceful postures, and symbolic hand gestures, which are deeply embedded in Sri Lankan cultural and religious traditions. Each movement, from footwork to upper body gestures, conveys meaning and emotion, often narrating stories from folklore or historical events. Mastering Udarata dance requires years of rigorous training, and even minor deviations in posture, timing, or coordination can compromise the aesthetic and cultural authenticity of the performance.

Traditionally, Udarata dance is taught in a teacher-led environment, where instructors provide continuous real-time correction and guidance. Students observe demonstrations, practice under supervision, and receive immediate feedback on their performance. While this apprenticeship model has ensured the transmission of cultural knowledge for generations, it also has inherent limitations. Students often need to practice independently, especially outside studio hours, and without constant teacher supervision, subtle errors in body alignment or gesture execution may go unnoticed and persist over time. Furthermore, the availability of skilled instructors is limited, particularly in rural or remote areas, making consistent guidance difficult to obtain.

In response to these challenges, motion tracking and computer vision technologies have begun to play a significant role in performing arts training globally. Pose estimation systems have been applied in fields such as Indian classical dance, Bharatanatyam, and Thai traditional dance to provide real-time feedback and performance analysis. These systems typically capture joint positions and movement trajectories to identify deviations from reference motions. By highlighting errors and suggesting corrections, they enable students to practice independently with higher accuracy.

However, the unique requirements of Udarata dance present specific challenges that existing systems do not fully address. Udarata dance involves rapid spins, intricate footwork, precise arm gestures, and subtle shifts in body weight, which standard motion tracking systems may not recognize accurately. Many conventional pose estimation models rely on datasets of general human movements, making them insufficient for capturing the dance-specific biomechanics required in Udarata. Therefore, there is a need for a specialized, accessible system that can accurately track and assess these movements in real time, enabling corrective feedback for students practicing alone.

This research proposes an interactive real-time motion tracking system for Udarata dance that utilizes MediaPipe, a framework developed by Google for real-time human pose estimation. MediaPipe uses lightweight models to detect body landmarks from standard video input, such as a webcam, without the need for specialized hardware or motion capture suits. By leveraging

MediaPipe, the system can track key joint positions shoulders, elbows, wrists, hips, knees, and ankles and compare the student's posture and movement to reference sequences performed by expert dancers. Deviations in posture, alignment, and timing are then identified and communicated as visual or audio feedback, guiding the student to correct their movements in real time.

The proposed system is designed to address several critical needs in Udarata dance training. First, it allows independent practice with accurate guidance, reducing the reliance on constant instructor supervision. Second, it provides instant feedback on subtle errors in body movement that may otherwise go unnoticed. Third, by logging performance data over time, it enables students and instructors to track improvement objectively. These capabilities not only enhance the learning experience but also support the preservation of Sri Lankan cultural heritage by ensuring that traditional dance movements are performed with precision and consistency.

From a technical perspective, the system combines computer vision, MediaPipe pose estimation algorithms, and motion analysis techniques. Video input from a webcam is processed in real time, extracting 3D joint coordinates that represent the student's posture and movement. Using a reference database of expert movements, the system calculates angles, distances, and spatial alignment of each joint to determine deviations from ideal performance. Feedback is delivered through an intuitive interface, enabling students to see exactly which movements need correction. Unlike traditional motion capture systems, this approach does not require expensive cameras or sensors, making it cost-effective and widely accessible for dance schools, community centers, and individual learners.

The research also has broader implications for cultural preservation and digital archiving. By digitizing Udarata dance movements and encoding them into a computational model, the system creates a digital record of expert performances that can be referenced for teaching, research, or archiving. This contributes to safeguarding intangible cultural heritage, ensuring that traditional dance forms are not lost or diluted over time due to modernization or reduced direct teacherstudent interaction. Moreover, the system can be extended to incorporate progress tracking, scoring, or gamification features, further enhancing motivation and engagement among students.

In addition, this research demonstrates how modern computational tools can complement traditional pedagogy, maintaining the authenticity of cultural practices while improving accessibility and learning efficiency. Unlike previous dance training technologies that required specialized equipment or were limited to general movements, this system is specifically tailored for Udarata dance, accounting for its unique movements, postures, and rhythm.

In conclusion, the background establishes the critical need for an accessible, real-time motion tracking system for Udarata dance. Traditional training methods, while culturally rich, face limitations in terms of independent practice, accuracy, and instructor availability. By leveraging MediaPipe for pose estimation and motion analysis, this research proposes a solution that enables precise self-guided learning, contributes to the preservation of cultural heritage, and introduces a technologically advanced, cost-effective tool for students and instructors alike. The project seeks to enhance the accuracy, consistency, and accessibility of Udarata dance education, creating a model that bridges cultural tradition and modern real-time motion tracking technology.

1.2 Literature Review

Traditional dance forms are increasingly recognized as forms of "living cultural heritage" (Tongpaeng et al., 2017). However, the preservation of these art forms particularly Thai, Korean, Bharatanatyam, and Sri Lankan Udarata dance faces significant challenges due to the gradual decline of master practitioners and the inherently subjective nature of the traditional Guru–Shishya (teacher–student) pedagogical model. Subramanian et al. (2025) argue that although visual observation has historically been the dominant teaching method, it lacks the consistency and objectivity required for independent and self-directed practice. Early digital preservation efforts attempted to address this issue. For instance, Tongpaeng et al. (2017) proposed a Thai dance framework that utilized Labanotation and 3D animation to document and archive movements. While such systems successfully created structured visual records, they remained largely non-interactive, underscoring the need for intelligent systems capable of delivering realtime corrective feedback to learners.

The technological landscape of dance analysis has evolved considerably, transitioning from expensive hardware-intensive systems to more accessible AI-driven frameworks. Early research predominantly relied on marker-based optical motion capture systems such as Vicon, as well as depth-sensing devices like Microsoft Kinect. Chan et al. (2011) developed a virtual reality (VR) learning environment in which students imitated a virtual instructor. Although the system demonstrated pedagogical effectiveness, the reliance on specialized suits and cameras limited its scalability and accessibility. In contrast, recent literature emphasizes markerless AI-based systems that use standard cameras. Maduwantha et al. (2024) critically examined the accessibility of motion capture technologies and concluded that software-based processing through conventional webcams significantly reduces the cost barrier for beginners. Within this domain, frameworks such as MediaPipe (BlazePose) and MoveNet have emerged as leading solutions. Kim (2025) demonstrated that BlazePose's 33-keypoint model offers superior suitability for traditional dance analysis because it captures detailed hand and foot placements that are absent in the standard 17-keypoint COCO model.

A central challenge in traditional dance analysis lies in differentiating between static postures and the dynamic flow of movements. Siriwardana and Maduranga (2024), in their study on Udarata dance, found that static pose detection methods are sufficient for validating foundational postures such as the Mandiya position but fail to capture the rhythmic and sequential complexity of movements like Goda Saramba. To address this limitation, researchers have increasingly adopted temporal modeling techniques using Long Short-Term Memory (LSTM) networks. Both Siriwardana and Maduranga (2024) and Kim (2025) employed Bi-directional LSTM (Bi-LSTM) models to process sequential pose data, achieving recognition accuracies exceeding 99% for traditional dance sequences. These approaches enable systems to evaluate not only individual frames but also the timing and transitions between movements. Additionally, dance movements often involve rapid spins and limb crossings that result in self-occlusion. Wang (2022) addressed this issue by introducing a lightweight neural network incorporating feature pyramids and hole convolution techniques to expand the receptive field, thereby maintaining pose stability even when body parts are partially hidden from the camera.

The effectiveness of AI-based dance learning systems ultimately depends on the clarity and objectivity of their feedback mechanisms. Saral et al. (2025) introduced the MoveMatch system, which calculates the Euclidean distance between a learner's detected keypoints and an expert reference model to generate a quantitative accuracy score. Similarly, Tongpaeng et al. (2018) developed a Thai Dance Training Tool that uses Position Accuracy Percentage (PAP) and Rotation Accuracy Percentage (RAP) metrics to assign standardized performance grades. In immersive environments, Chan et al. (2011) implemented a "Super Mirror" concept in VR, where body segments change color yellow indicating correctness and red signaling errors to guide learners visually. Expanding on immersive evaluation, Esaki and Nagao (2024) developed a reference-guided VR model capable of predicting performance ratings on a 10-point scale, aligning AI-based feedback with human coaching standards. Furthermore, inclusivity has emerged as a critical dimension in dance technology research. Amarasooriya (2023) adapted MediaPipe-based systems for hearing-impaired dancers by replacing auditory rhythmic cues with real-time visual feedback, demonstrating how AI technologies can democratize access to traditional dance education.

Despite significant advancements in AI-driven dance analysis, several research gaps remain.
Cultural specificity remains a major limitation, as most systems are trained on Western or East Asian movement datasets. The unique biomechanics of Kandyan Udarata dance such as the lowcentered Mandiya posture and rhythmically intensive foot-stamping patterns require specialized datasets tailored to local traditions. Additionally, while real-time feedback is a primary objective, high-accuracy deep learning models often introduce latency or depend on high-end GPU hardware. Wang and Ngoi (2023) highlight the need for optimized MediaPipe-based frameworks capable of operating efficiently in low-resource environments, such as standard laptops with webcams, without compromising accuracy. Finally, existing systems largely focus on measurable joint angles and positional accuracy while overlooking expressive qualities such as fluidity, symbolic gestures, and emotional articulation. Although Esaki and Nagao (2024) begin addressing automated performance evaluation beyond basic joint metrics, applying such holistic assessment methods to the expressive dimensions of Sri Lankan traditional dance remains an open research challenge.

1.3 Research Problem

Udarata dance is a valuable form of intangible cultural heritage that has been transmitted from generation to generation through traditional teaching practices. The learning process primarily follows a teacher-centered, face-to-face approach, where knowledge is transferred directly from the guru to the student. In a typical classroom setting, students observe the teacher's posture and movements, imitate them, and receive immediate corrections to improve the accuracy and precision of their performance.

However, achieving mastery in Udarata dance requires consistent individual practice beyond classroom sessions. During self-practice, students do not have access to real-time guidance or corrective feedback from their teacher. As a result, inaccuracies in posture, hand gestures, footwork, and rhythmic movements may go unnoticed and become habitual over time. These uncorrected errors can negatively affect the technical quality, authenticity, and preservation of the dance form.

Therefore, there is a need for a system that can provide objective, real-time feedback to learners during independent practice, ensuring movement accuracy while supporting the preservation of Udarata dance heritage.

1.4 Research Gap

An identifiable research gap exists when comparing the proposed system with currently available motion analysis solutions applied to traditional dance. While motion tracking technologies have been widely used in ballet, contemporary dance, and Indian classical dance forms such as Bharatanatyam, research specifically focusing on Udarata (Kandyan) dance remains very limited. Most existing Sri Lankan studies primarily focus on static pose detection or isolated movements, rather than evaluating complete rhythmic movement sequences. This approach does not adequately capture the temporal flow, rhythm, and transitions between movements, which are essential components of Kandyan dance performance.

From a technical perspective, there is a lack of a structured computational framework that translates the qualitative biomechanical rules of Kandyan dance into quantitative mathematical constraints. Generic motion analysis systems and commercial dance-learning platforms are typically designed for general human movement or fitness applications and do not account for the specific biomechanical requirements of Udarata dance, such as the precise knee flexion depth in the Mandiya posture, symmetrical arm elevations, controlled torso alignment, and rhythmic foot stamping patterns. Without incorporating such expert-defined movement rules, existing systems cannot provide accurate and culturally appropriate corrective feedback for learners.

Furthermore, the research domain suffers from the absence of standardized datasets for Udarata dance movements. Current studies often rely on raw video recordings that are sensitive to camera distance, dancer positioning, and environmental variations. There is a clear lack of normalized three-dimensional landmark datasets that can serve as reliable ground truth references for movement comparison. Without position-invariant data representations, it becomes difficult to build systems that perform consistently across different users, camera setups, and practice environments.

Finally, there is a significant gap in developing a fully integrated system that combines markerless pose tracking, dynamic movement sequence analysis, and real-time corrective feedback specifically tailored for Kandyan dance training. Most currently available learning methods remain passive, relying on video demonstrations or instructor supervision without providing immediate automated feedback during independent practice. Therefore, a clear research gap exists in creating an accessible, real-time motion analysis system that supports effective self-learning while preserving the biomechanical authenticity of Udarata dance movements.

Novelty of the Research (Udarata Version)

The novelty of this research lies in the development of a specialized motion analysis framework designed specifically for Udarata (Kandyan) dance biomechanics. Unlike existing general-purpose pose tracking systems, this research introduces a methodology that converts qualitative dance knowledge provided by expert Kandyan dancers into quantitative biomechanical parameters. Through this approach, traditional movement rules such as the Mandiya knee posture, arm symmetry, shoulder alignment, and rhythmic stamping patterns can be evaluated using computational models derived directly from expert performance data.

Another innovative contribution is the implementation of a coordinate-based motion representation using Hip-Center Normalization. This technique ensures that the recorded dance movements become independent of the dancer’s height, camera position, or distance from the camera, allowing the system to analyze movement patterns based on normalized spatial trajectories rather than raw pixel data. By transforming a standard RGB webcam into a markerless motion capture sensor, the system extracts 33 three-dimensional body landmarks using MediaPipe Pose.

In addition, the system employs Dynamic Time Warping (DTW) to compare a learner’s movement sequences with expert reference performances, enabling accurate similarity evaluation even when movements are performed at different speeds. This capability allows the system to analyze both postural accuracy and rhythmic movement flow, which are critical characteristics of Kandyan dance.

By integrating pose estimation, biomechanical validation rules, and real-time feedback mechanisms, the proposed system introduces a novel, accessible, and culturally informed digital learning platform for Udarata dance training. Moreover, the creation of a structured digital archive of expert Kandyan dance movements contributes to the long-term preservation and documentation of Sri Lanka’s intangible cultural heritage while supporting modern technology-assisted dance education.

2.RESEARCH OBJECTIVES

2.1 Main Objective

To develop and evaluate an integrated real-time motion analysis system for Udarata (Kandyan) dance that supports accurate and independent practice through automated posture analysis and real-time corrective feedback.

2.2 Specific Objectives

1. **Expert Modeling and Archiving**
   To develop a digital reference library of foundational Udarata dance movements by recording expert Kandyan dancers and extracting normalized 3D pose trajectories for core techniques such as *Mandiya posture, Vattam rotations, arm gestures, and rhythmic stamping patterns*. These expert sequences will serve as the ground truth dataset for system evaluation.

2. **Markerless Motion Tracking**
   To implement a high-precision pose tracking system using MediaPipe Pose capable of capturing 33 three-dimensional body landmarks from live video input using a standard RGB webcam.

3. **Temporal Sequence Evaluation**
   To integrate a dynamic movement analysis engine using the Dynamic Time Warping (DTW) algorithm to evaluate the rhythm, flow, and temporal transitions of continuous Udarata dance movements, regardless of differences in the learner's performance speed.

4. **Real-Time Corrective Feedback**
   To design a real-time feedback mechanism that provides quantitative similarity scores (0–100%) and biomechanical guidance such as knee flexion depth, arm symmetry, shoulder alignment, and posture corrections, with feedback latency below 100 milliseconds.

5. **Statistical Validation of Improvement**
   To evaluate the system's effectiveness through experimental testing with Udarata dance learners, measuring improvements in movement accuracy through pre-test and post-test comparisons against traditional video-based learning methods.

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science Research (DSR) methodology, as the study focuses on designing, developing, and evaluating a novel technological artifact, an interactive motion tracking system for Udarata (Kandyan) dance learning.

The Design Science approach is suitable because the research aims to develop a practical technological solution that addresses a real-world problem in dance training and cultural preservation.

The research process consists of the following stages:

1. **Problem Identification** – Identifying limitations in independent practice of Udarata (Kandyan) dance, particularly the lack of real-time corrective feedback during self-practice.

2. **Requirement Analysis** – Defining functional, technical, and biomechanical requirements based on Kandyan dance fundamentals such as Mandiya posture, arm alignment, rhythmic stamping patterns, and torso positioning.

3. **Artifact Development** – Designing and implementing the motion tracking system, including pose detection, movement sequence comparison, and feedback generation mechanisms.

4. **Evaluation** – Measuring system performance in terms of pose similarity accuracy, processing latency, and improvement in learner movement accuracy.

5. **Validation** – Testing the system with real users to evaluate usability and effectiveness for supporting independent practice of Kandyan dance movements.

This methodology ensures both technical innovation and practical usability in the development of the proposed system.

3.2 Tools and Technologies

- MediaPipe = Real-time 3D landmark detection.
- Python = Core development language.
- OpenCV = Video frame processing.
- NumPy & Pandas = Data processing and normalization.
- Dynamic Time Warping (DTW) = Sequence similarity analysis.

- TensorFlow (Optional Extension) = For future deep learning enhancements.

These tools were selected due to their low hardware requirements, open-source availability, and compatibility with standard webcams.

3.3 Data Collection and Ethical Considerations

Data will be collected by recording expert Udarata dancers performing selected foundational sequences. Written consent will be obtained before recording.

All video recordings will be used strictly for research and model development purposes. Personal identity information will not be publicly disclosed. Data will be stored securely with restricted access.

Students participating in system evaluation will also provide informed consent. The system will comply with data privacy and ethical research standards.

3.4 Validation Strategy and Performance Metrics

The effectiveness of the proposed system will be evaluated using an experimental testing approach involving Kandyan dance learners.

Statistical Validation

Participants will be divided into two groups:

- **Control Group** - Learners practicing using traditional video-based learning materials.
- **Experimental Group** - Learners practicing using the proposed real-time motion tracking system.

Learner performance will be evaluated by measuring the Postural Error Rate, calculated as the Euclidean distance between the learner's detected joint positions and the expert reference templates. A paired t-test statistical analysis will be conducted to determine whether the improvement in movement accuracy between the two groups is statistically significant.

Performance Metrics

The system will also be evaluated using the following technical metrics:

- **Pose Similarity Accuracy** - Degree of similarity between learner movements and expert reference sequences.

- **System Latency** - Feedback delay measured in milliseconds to ensure real-time interaction.
- **Frame Processing Rate** - Maintaining approximately **30–60 frames per second (FPS)** for smooth motion tracking on standard consumer laptops.
- **System Stability** - Ability to run continuously during practice sessions without performance degradation.

These evaluation metrics ensure that the system meets both technical performance standards and educational effectiveness requirements.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview

The proposed solution is a real-time motion analysis and tutoring system designed specifically for the biomechanical analysis of Udarata (Kandyan) dance movements. The system processes raw video input captured through a standard RGB webcam and converts it into normalized mathematical motion trajectories that can be compared with expert reference movements.

By separating the video acquisition layer from the dance-specific motion analysis logic, the architecture ensures a scalable, modular, and hardware-agnostic system. This approach allows accurate skeletal motion analysis using only a standard webcam, eliminating the need for specialized motion capture equipment.

The architecture is designed to provide real-time feedback to learners practicing Kandyan dance movements, enabling them to identify posture deviations and correct their movements during independent practice sessions.

4.2 System Diagram

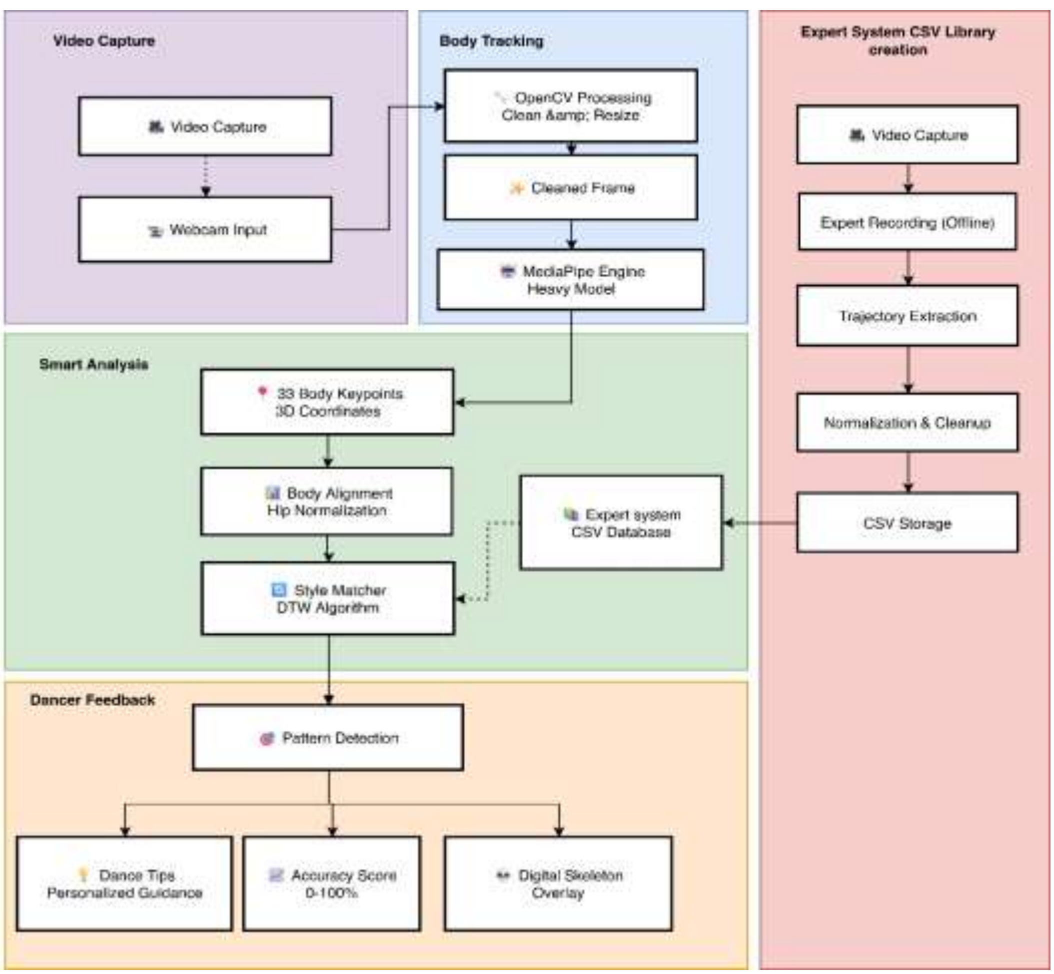

Figure 1 - System Diagram

4.3 Individual Components and Their Functions

The system architecture is divided into four primary tiers, each responsible for a specific stage in the motion analysis pipeline.

1. Video Capture Layer

This layer is responsible for capturing the real-time RGB video stream from the user's webcam. The captured frames serve as the raw visual input for the motion analysis process.

2. Body Tracking Layer

The body tracking layer performs frame preprocessing and human pose detection. OpenCV is used to handle video frame processing tasks such as resizing and image optimization. The processed frames are then passed to the MediaPipe Pose engine, which detects and extracts 33 three-dimensional body landmarks representing key joint positions of the dancer.

These landmarks include critical joints such as shoulders, elbows, wrists, hips, knees, and ankles, which are essential for analyzing Kandyan dance postures and movements.

3. Motion Analysis Layer

This layer represents the core computational component of the system. It performs motion analysis by converting the detected landmarks into normalized motion data and comparing them with expert reference movements.

First, Hip-Center Normalization is applied to all landmark coordinates. This process recalculates the positions of body landmarks relative to the midpoint between the left and right hip joints, ensuring that movement analysis remains independent of the dancer’s height, position, or camera distance.

Next, the system uses a Dynamic Time Warping (DTW) algorithm to compare the learner’s motion sequence with pre-recorded expert Kandyan dance reference sequences stored in a structured CSV database. DTW enables accurate similarity evaluation even when the learner performs movements at a slightly different speed from the expert reference.

4. Feedback and Visualization Layer

The final layer is responsible for generating real-time feedback and visual guidance for the learner.

This layer performs movement pattern evaluation and produces the following outputs:

- Similarity score (0–100%) representing how closely the learner’s movement matches the expert reference.
- Personalized corrective guidance, such as suggestions to adjust knee bend, arm alignment, or posture.

- Digital skeletal overlay visualization, allowing users to see their tracked body posture in real time.

This feedback helps learners identify movement errors and improve their performance during practice.

4.4 Component Interactions

The system operates through a **sequential data processing pipeline**.

The **Video Capture layer** continuously sends raw video frames to the Body Tracking layer, where visual data is converted into numerical pose landmarks using the MediaPipe pose estimation model.

These landmark coordinates are then transmitted to the Motion Analysis layer, where normalization and sequence matching algorithms analyze the learner's movements. This layer retrieves reference data from the Expert Movement Database, which is created during an offline data collection phase using expert Kandyan dancers.

Finally, the analyzed results are passed to the Feedback and Visualization layer, which updates the user interface with movement scores, corrective guidance, and real-time pose visualization.

4.5 Data Flow Explanation

The system processes a continuous RGB video stream at approximately 30 frames per second (FPS).

First, OpenCV performs preprocessing operations such as frame resizing and image optimization before the frames are passed to the MediaPipe Pose inference engine.

MediaPipe extracts 33 three-dimensional body landmarks (x, y, z coordinates) representing the dancer's skeletal structure. These landmarks are then transformed using hip-centered coordinate normalization, which eliminates variations caused by camera distance, body height, or user positioning.

A temporal sliding window buffer stores the most recent 60 frames of normalized pose data, enabling the system to analyze movement sequences rather than individual static poses.

The buffered pose data is then processed by the Dynamic Time Warping (DTW) algorithm, which calculates the similarity between the learner's motion sequence and the expert reference template. Based on this similarity score and biomechanical validation rules, the system generates real-time feedback and movement correction guidance.

4.6 Development Procedure

The development process follows a structured pipeline that transforms raw video input into biomechanical analysis and real-time corrective feedback tailored specifically for Udarata (Kandyan) dance.

Phase 1- Expert Modeling and Reference Database (Offline Phase)

Ethogram Creation

Structured movement definitions (ethograms) are created based on Udarata dance fundamentals. These include core postures and movement sequences such as *Mandiya* controlled arm elevations, rhythmic foot stamping patterns, and torso alignment rules. The biomechanical constraints of Kandyan dance such as knee flexion depth, symmetrical arm extension, and upright spine positioning are formally defined as reference parameters.

Expert Recording

A certified Udarata dance practitioner performs selected foundational sequences. Movements are recorded using a standard RGB webcam to ensure accessibility and real-world applicability.

Coordinate Extraction

The MediaPipe Pose framework extracts 33 three-dimensional body landmarks (x, y, z coordinates) for every frame of the expert's performance.

Data Normalization

To ensure position invariance and eliminate spatial bias, a Hip-Center Normalization algorithm is applied. All landmark coordinates are re-centered using the midpoint between the left and right hip landmarks (Landmarks 23 and 24). This ensures that comparisons are independent of camera placement and dancer positioning.

Reference Library Generation

The normalized landmark trajectories are stored in structured CSV format, forming a Master Udarata Dance Reference Library. This dataset acts as the biomechanical ground truth for realtime comparison during student practice.

Phase 2- Live Tutoring and Recognition Engine (Online Phase)

Real-Time Tracking

The system captures the student's live video feed and extracts 3D pose landmarks in real time using MediaPipe.

Temporal Sliding Window
Instead of analyzing individual static poses, the system stores a sliding window of the most recent 60 frames of normalized landmark data. This enables evaluation of movement rhythm, transition smoothness, and dynamic flow characteristic of Udarata dance.

Pattern Matching using Dynamic Time Warping (DTW)
A Dynamic Time Warping (DTW) algorithm compares the student's live motion window with the expert reference sequences stored in the Master Library. DTW allows accurate similarity scoring even when the student performs movements at slightly different speeds.

Biomechanical Validation
A rule-based validation layer checks Udarata-specific posture constraints, including:

- Knee flexion angle depth in *Mandiya*
- Symmetry of arm elevation and shoulder alignment
- Spine verticality during stamping sequences
- Relative positioning of hands with respect to head and torso

Deviation thresholds are applied to detect incorrect execution.

Phase 3- Feedback Loop and System Optimization

Engagement Estimation
Optional telemetry indicators such as head stability, repetition frequency, and hesitation patterns are analyzed to estimate student focus and engagement levels during practice sessions.

Feedback Generation
Personalized corrective prompts are displayed via the user interface (e.g., "Increase Knee Bend," "Raise Left Arm," "Align Shoulders").

Performance Optimization
The system is optimized to maintain smooth real-time performance with minimal latency (target <100ms). Efficient computation ensures compatibility with standard laptops and webcams without requiring specialized hardware.

4.7 Scalability and Security

The system is lightweight and can run on standard laptops. Cloud integration can be added in future versions for large-scale deployment. User data will be encrypted and stored securely.

5. USER REQUIREMENTS

Key Stakeholders

- Udarata dance learners (beginners, intermediate students, rural practitioners)
- Certified Kandyan dance gurus for validation
- Cultural institutions focused on heritage preservation
- Researchers analyzing traditional movement biomechanics

5.1 Functional Requirements

1. The system shall capture real-time learner movements using a webcam or smartphone camera during Udarata practice sessions.
2. The system shall extract 33 body landmarks using MediaPipe Pose to analyze footwork, arm gestures, and posture alignment specific to Kandyan dance.
3. The system shall compare learner movements against expert Udarata reference sequences stored in the database.
4. The system shall detect posture deviations based on predefined Udarata biomechanical rules.
5. The system shall highlight incorrect joints and body segments using real-time visual overlays.
6. The system shall generate automated performance reports including similarity scores and posture correction suggestions.
7. The system shall support structured lesson progression for foundational Udarata techniques such as Mandiya posture, Vattam sequences, and rhythmic stamping patterns.

5.2 Non-Functional Requirements

Performance

- Real-time feedback latency shall be less than 100 milliseconds.
- The system shall achieve a minimum of 90% pose similarity accuracy under controlled testing conditions.

Usability

- The interface shall be intuitive and beginner-friendly.
- The system shall support multilingual guidance if required.

Accessibility

- The system shall function on standard laptops with webcams.
- No specialized motion capture hardware shall be required.

Security

- User consent shall be obtained before recording practice sessions.
- Performance data shall be stored securely with privacy protection measures.

Reliability

- The system shall operate continuously for at least 30-minute practice sessions without crashes.
- Practice progress shall be automatically saved.

6. COMMERCIALIZATION PLAN

The proposed Udarata (Kandyan) Dance Learning System has strong potential for commercialization as a technology-assisted training platform for traditional dance education in Sri Lanka. The system provides real-time motion tracking and automated feedback, which can support students practicing outside traditional classroom environments.

Initially, the platform will focus specifically on Udarata (Kandyan) dance, one of the most widely practiced classical dance traditions in Sri Lanka. According to cultural education statistics, thousands of students learn Kandyan dance through dance academies, cultural institutions, and school programs, representing a significant potential user base.

The system can later be expanded to support other Sri Lankan dance traditions such as Pahatharata (Low Country), Sabaragamuwa, and Jana Natum, enabling the development of a comprehensive digital learning platform for Sri Lankan traditional dance.

Target Market

The primary target users of the system include:

- Dance academies in Sri Lanka
- Cultural training institutes
- School dance programs
- Individual dance learners
- Online dance education platforms
- Cultural tourism and performance training centers

Value Proposition

The proposed system offers several unique benefits compared to traditional dance learning methods:

- Provides real-time corrective feedback during independent practice.
- Enables objective evaluation of dance movements using pose similarity analysis.
- Supports self-paced learning outside classroom sessions.
- Contributes to the digital preservation of Kandyan dance movements through expert motion archiving.

Revenue Model

The system can be commercialized using a freemium or subscription-based model.
Possible pricing strategies include:

- Free Version - Basic practice features and limited movement analysis.
- Premium Individual Subscription - Monthly access to advanced feedback, full movement libraries, and performance reports.
- Institutional Licensing - Dance academies and cultural institutions can purchase multi-user access for their students.

Competitive Advantage

The proposed system offers several advantages over existing dance learning tools:

- Specifically designed for Udarata (Kandyan) dance biomechanics
- Markerless motion tracking using standard webcams
- Low hardware requirements with no specialized motion capture equipment
- Real-time corrective feedback during practice sessions
- Accessible and affordable for individual learners and institutions

Intellectual Property

The software architecture, motion analysis algorithms, and curated dance reference datasets developed through this research can be protected under software copyright laws. Additionally, the expert movement database of Kandyan dance sequences may be considered a valuable digital cultural archive that can support both educational and research applications.

7. BUDGET AND JUSTIFICATION

Table 1 - Budget

Category	**Cost (LKR)**	**Justification**
Hardware	**6,000**	Tripods only (using laptop webcams)
Software	**0**	All open-source (MediaPipe, Python)
Human Effort	**40,000**	**students + 4 teachers**
Travel/Other	**14,000**	Travel + printing
TOTAL	**60,000**	Final Year project

The total estimated budget for this research component is 60,000 LKR, structured to maximize the use of cost-effective, open-source technologies while prioritizing data quality. A significant portion of the funds (40,000 LKR) is allocated toward Human Effort, which includes honorariums for the four expert Sabaragamuwa dance teachers required to create the Expert System CSV Library and incentives for the 60 students participating in the statistical validation phase.

By leveraging the MediaPipe Tasks API and existing laptop webcams, software costs have been eliminated, and hardware expenses are limited to essential equipment like tripods to ensure stable video capture. The remaining funds cover travel for field data collection and the printing of research materials, ensuring the project remains within a feasible undergraduate scope while maintaining high standards of biomechanical accuracy and cultural preservation.

8. WORK BREAKDOWN STRUCTURE (WBS)

The project is divided into structured phases to ensure systematic development, implementation, and evaluation of the Udarata dance motion tracking system.

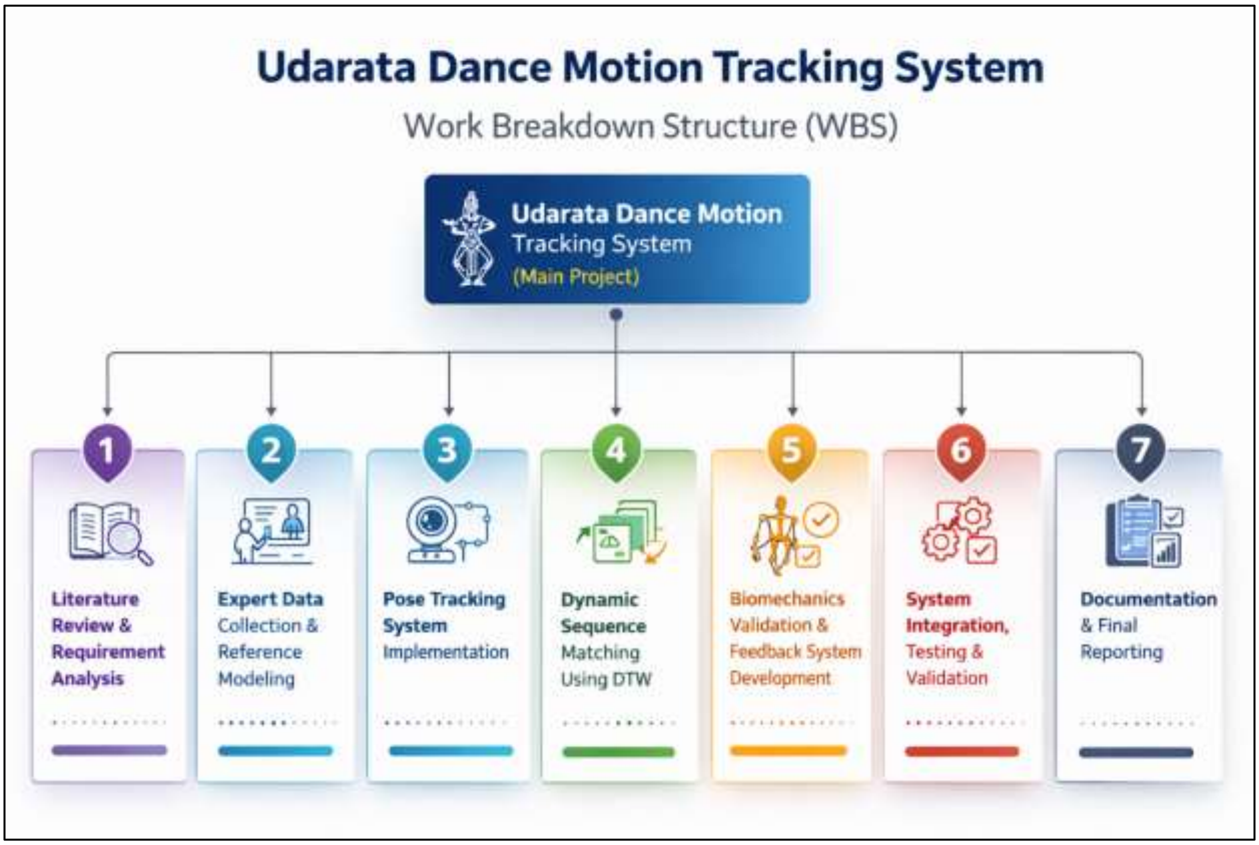

Figure 2 - Work Flow diagram

1. Literature Review and Requirement Analysis

- Review scholarly research on pose estimation, MediaPipe, motion tracking, and dance biomechanics.
- Analyze existing dance training technologies and identify technical limitations.
- Study biomechanical characteristics of Udarata (Kandyan) dance including Mandiya posture, Vattam sequences, and rhythmic stamping.
- Define functional and non-functional system requirements.

- Identify performance benchmarks (accuracy, latency, stability).
- Prepare proposal documentation and initial design specifications.

Deliverable: Requirement Specification Document and Research Foundation.

2. Expert Data Collection and Reference Modeling

- Identify and collaborate with a certified Udarata dance practitioner.
- Record foundational dance movements using a standard RGB webcam.
- Select key movement sequences for modeling (e.g., Mandiya, Vattam, arm gestures).
- Extract 3D pose landmarks using MediaPipe.
- Apply hip-centered normalization to eliminate spatial bias.
- Store processed sequences as structured CSV reference datasets.

Deliverable: Master Udarata Dance Reference Library (Ground Truth Dataset).

3. Pose Tracking System Implementation

- Integrate MediaPipe Pose framework into the development environment.
- Develop real-time video capture and landmark extraction pipeline.
- Implement coordinate normalization algorithm.
- Optimize landmark detection for consistent frame processing.
- Ensure system stability under continuous real-time capture.

Deliverable: Functional Real-Time Pose Tracking Module.

4. Dynamic Sequence Matching using DTW

- Implement Temporal Sliding Window mechanism (60-frame buffer).
- Develop Dynamic Time Warping (DTW) algorithm for sequence comparison.
- Define similarity scoring mechanism.
- Set threshold values for acceptable vs incorrect movement execution.
- Test algorithm using recorded expert sequences and sample learner data.

Deliverable: Motion Similarity Evaluation Engine.

5. Biomechanics Validation and Feedback System Development

- Define Udarata-specific biomechanical constraints:
    - Knee angle depth
    - Arm symmetry
    - Shoulder alignment
    - Spine posture
- Develop rule-based validation layer.

- Design user interface for corrective feedback messages.
- Integrate similarity score display and error highlighting.

Deliverable: Real-Time Corrective Feedback System.

6. System Integration, Testing, and Validation

- Integrate all modules:
    - Video Capture
    - Pose Estimation
    - DTW Engine
    - Biomechanics Validator
    - UI Feedback Layer
- Conduct unit testing and integration testing.
- Measure performance metrics:
    - Latency
    - Pose similarity accuracy
    - Frame stability
- Conduct pilot testing with Udarata learners.
- Collect pre-test and post-test performance data.
- Analyze improvement rate statistically.

Deliverable: Validated and Performance-Optimized System.

7. Documentation and Final Reporting

- Document system architecture and algorithms.
- Compile methodology and evaluation findings.
- Prepare final dissertation report.
- Create presentation slides and demonstration materials.
- Conduct final system demonstration and viva preparation.

Deliverable: Final Dissertation and Presentation.

9. GANTT CHART

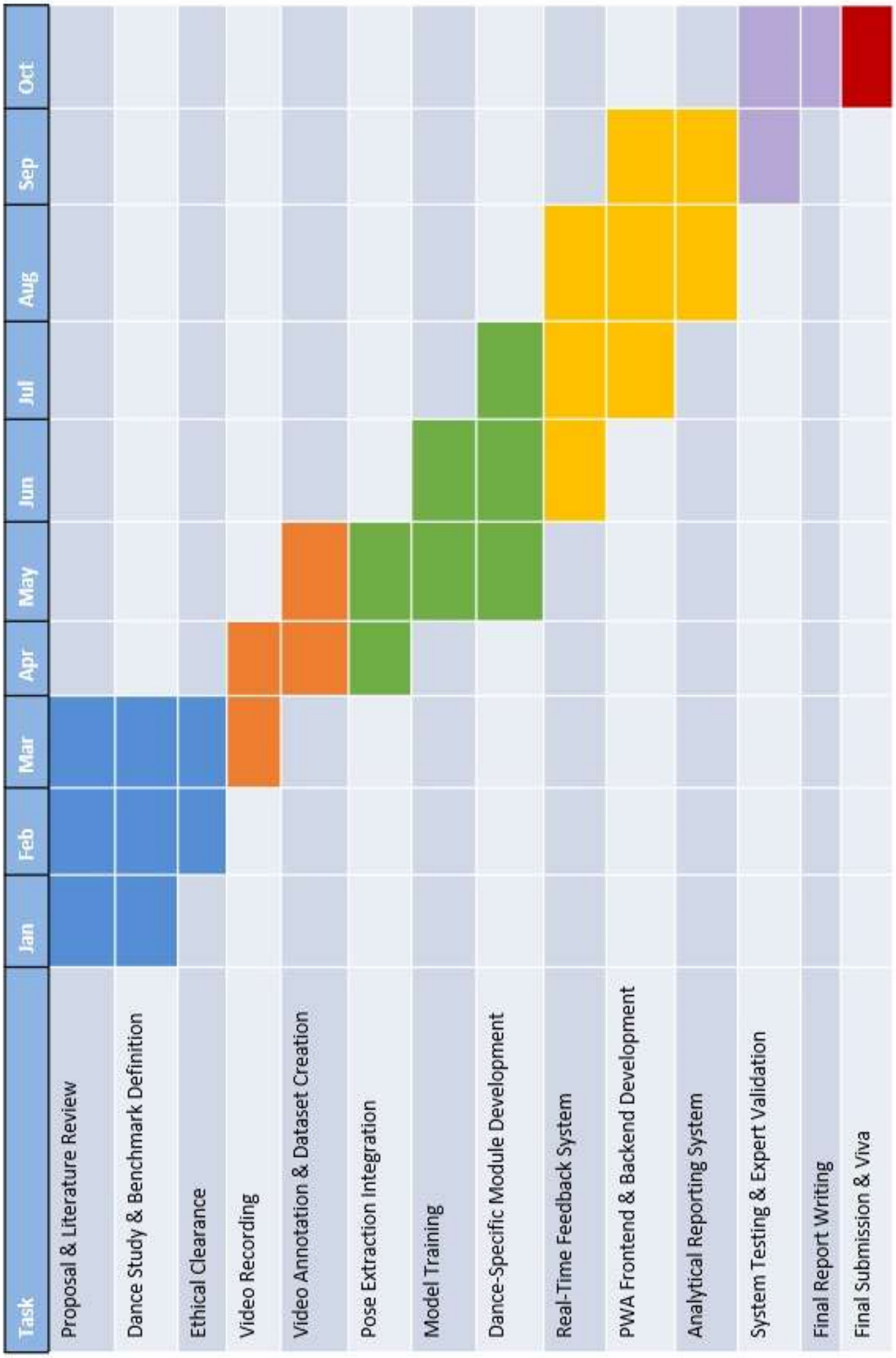

Figure 3 - Gran chart

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